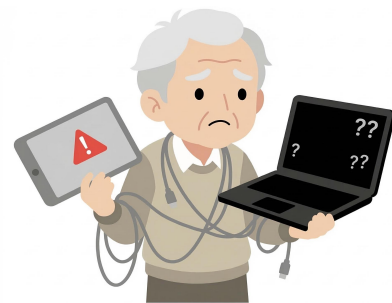
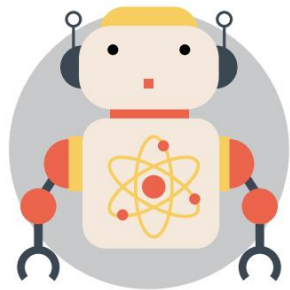


基于大模型智能体的老年人数字抵触形成 与演进机制研究

从数字摩擦到习得性无助的
社会计算



项目背景

数字摩擦对老年人信心的摧毁与习得性无助现象



01

数字摩擦的累积效应 —

交互挫败不断累积，导致老年人产生不可控感。

02

自身消极的归因偏差 —

倾向于内在、稳定、全局的解释风格

将可能的设计缺陷错误归结为“老了、笨、时代变了”。

03

负面预期的心理闭环 —

确信努力也无法改变未来结果

04

产生习得性无助 —

How older people account for their experiences with interactive technology

289

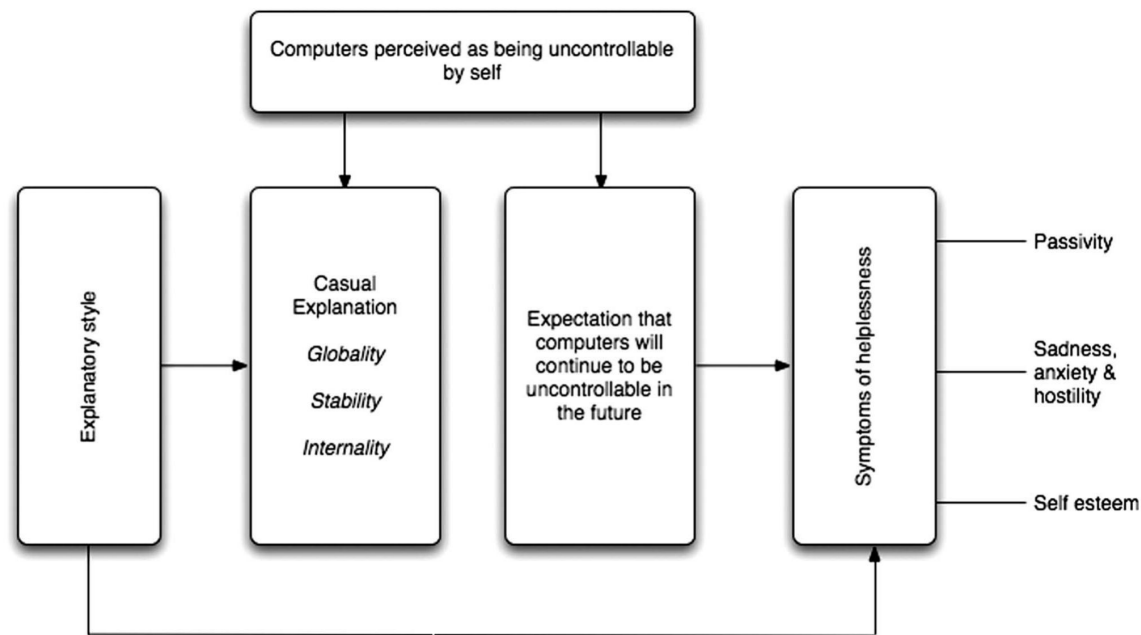


Figure 1. Explanatory style, causal explanations and learned helplessness in relation to learning to use computers.

Turner, Phil & Turner, Susan & Van de Walle, Guy. (2007). How older people account for their experiences with interactive technology. *Behaviour & Information Technology*. 26. 287–296. 10.1080/01449290601173499.



项目背景

传统研究方法的局限性

01 | 现实社会实验的困境

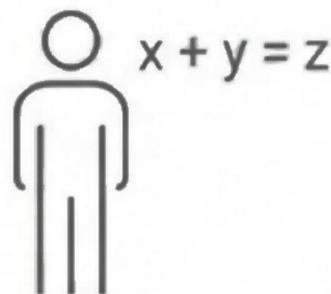


极高的成本与时间跨度



伦理与道德上的风险

02 | 传统Agent(ABM)的瓶颈 Underdeveloped Intelligence



基于规则、方程或机器学习模型驱动。

行为模式单一，缺乏人类认知的复杂性。

难以产生用于深层机制解释的定性数据。

项目背景

社会模拟计算的优势与可行性

01 | 模拟社会实验的优势



较低的成本与时间跨度



不存在伦理与道德上的风险

可展现出类似人类的非理性、情感决策。

02 | LLM-Based Agent的优势



可以提供更多的数据

Agent具备可被追问的心智，用于机制解释。

已有社会模拟平台的比较

Table 1: Comparison of LLM-driven agents and social simulators.

Model	Minds			Mobility		Economics		Social		Others		M-B		Scale	Env.
	RP.	AM.	SI.	Relo.	Traj.	Res.	Exc.	Int.	Rel.	Con.	Free	Infl.	Dri.	#	
D2A [101]	✓	✓	✓	✓						✓	✓			<100	Text
Ecoagent [64]	✓	✓				✓	✓					✓		100-1k	Rules
OASIS [109]	✓	✓						✓	✓			✓		>10k	Rules
GA1000 [78]	✓	✓	✓											1k-10k	×
MATRIX [75]	✓	✓						✓						<100	Text
Sociodojo [24]	✓					✓								<100	Data
GA [77]	✓	✓		✓	✓			✓	✓	✓	✓	✓	✓	<100	Rules
GenSim [94]	✓	✓								✓				>10k	Data
Project Sid [6]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	1k-10k	Rules
AgentScope [40]	✓	✓						✓						N/A	N/A
HiSim [72]	✓	✓						✓	✓					100-1k	Rules
S3	✓	✓						✓	✓					1k-10k	Rules
Agent4Rec [110]	✓	✓								✓				1k-10k	Rules
RecAgent [100]	✓	✓						✓	✓	✓				100-1k	Rules
Sotopia [116]	✓							✓						<100	×
Casevo [52]	✓	✓						✓	✓					100-1k	Rules
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	>10k	Society

Piao, Jinghua & Yan, Yuwei & Zhang, Jun & Li, Nian & Yan, Junbo & Lan, Xiaochong & Lu, Zhihong & Zheng, Zhiheng & Wang, Jing Yi & Zhou, Di & Gao, Chen & Xu, Fengli & Zhang, Fang & Rong, Ke & Su, Jun & Li, Yong. (2025). AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society. arXiv. 2502.08691. <https://arxiv.org/abs/2502.08691>.

AgentSociety的优势

Silicon Participants

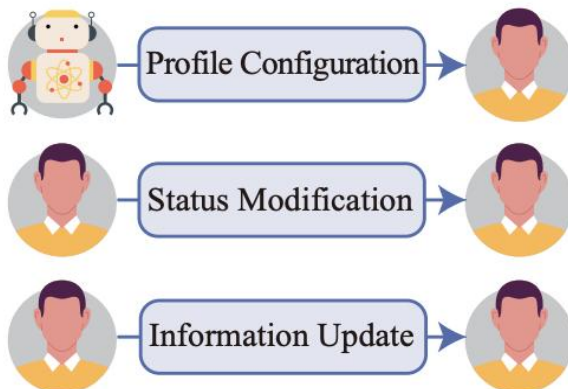


Profiles & Status

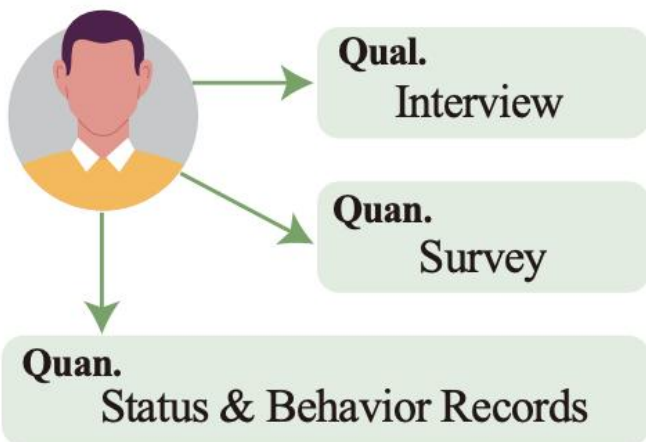
Memory & Minds

Social Behaviors

Interventions



Data Collection



01

Agent层面

LLM赋予了agent响应能力,可测量性,可追问性

02

实验流程上

提供了一套完整的实验框架, 包含
Silicon Participants;
干预机制; 完整的数据收集工具链

03

基于社会科学理论的建模

保证了agent可解释性和可控性

研究空白

01 | 现有研究已经发现了老年人的科技焦虑、自我效能感、感知有用性社会支持等会显著影响其数字使用意愿与恐惧。

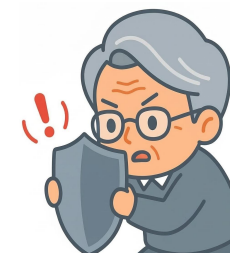
但多数研究仍以横断面调查、结构方程或定性访谈为主，主要回答哪些因素相关，而较少解释老年人如何在连续数字摩擦中逐步形成数字抵触与习得性无助。

02 | 现有方法以问卷、访谈和短期实验为主，难以在伦理可控、成本可承受的条件下持续追踪老年人数字抵触的长期演进过程。

且传统ABM虽可进行社会模拟，但对归因偏差、情绪波动、求助与回避等认知心理过程刻画不足，难以支撑形成机制层面的解释。

03 | 现有干预研究已表明部分支持能够改善老年人的自我效能感与使用意愿等
与此同时，silicon participants用于 pilot social experiments 的可行性也已被验证。
但目前两类研究尚未真正结合，缺少老年人数字抵触问题的理论驱动仿真框架与干预评估平台。

项目设置Agent异质性建模

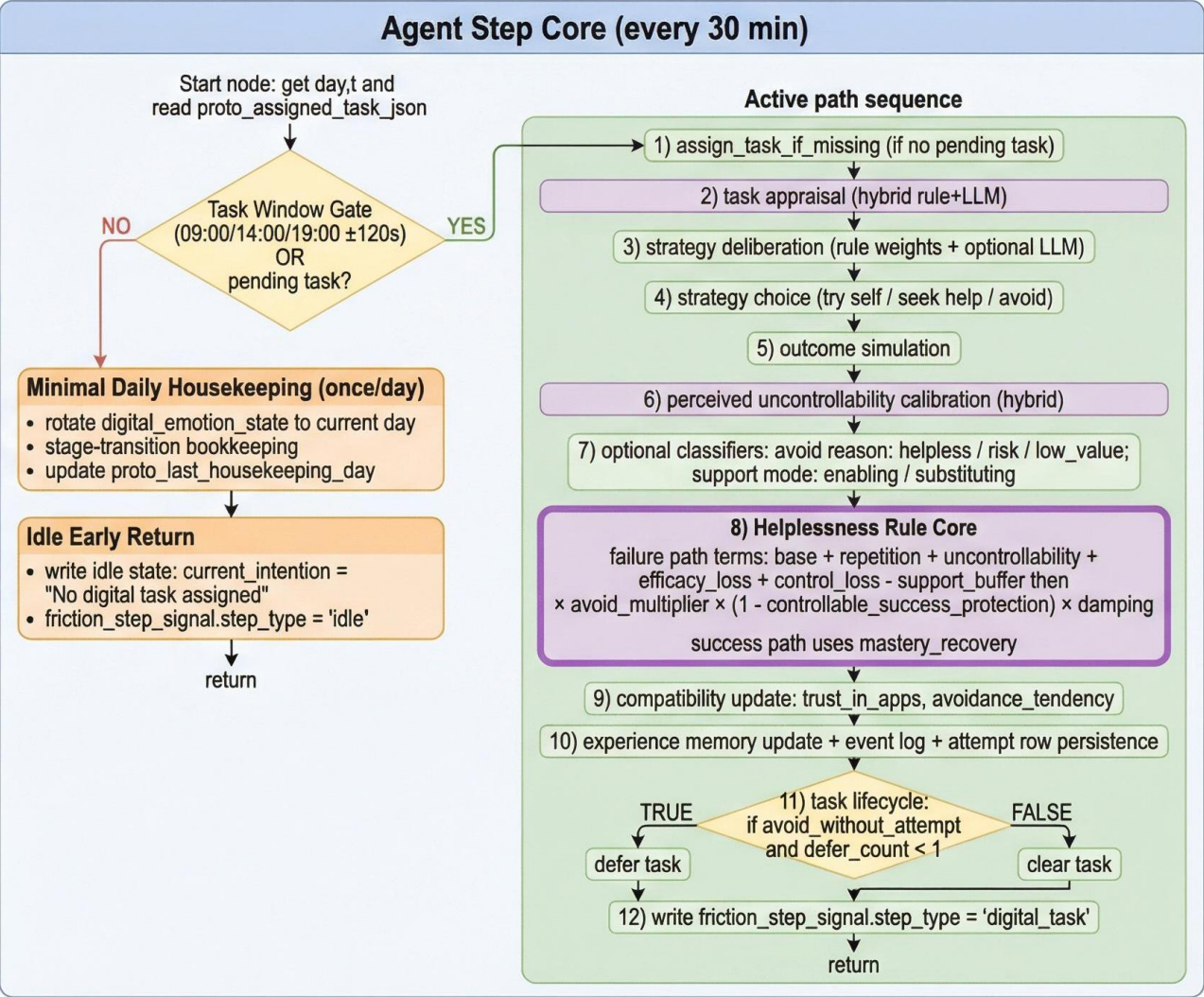
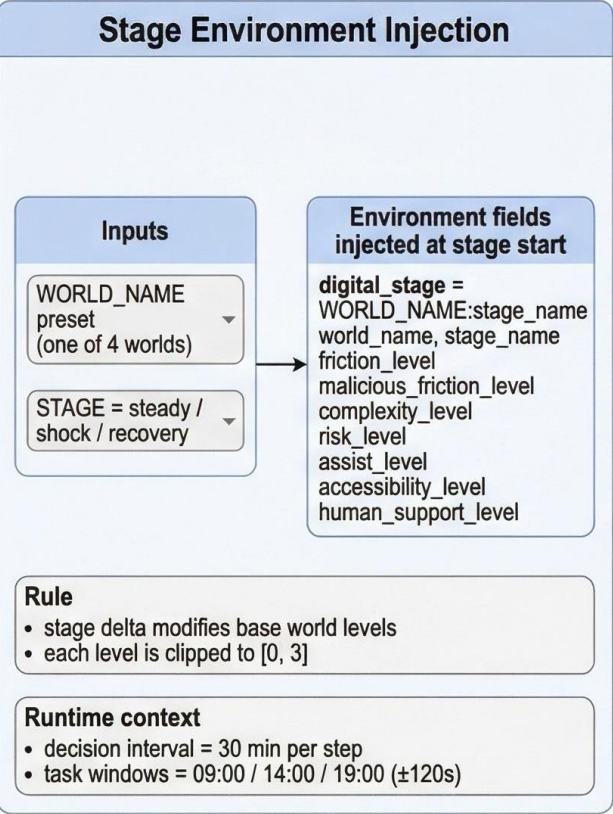


项目设置Agent异质性建模

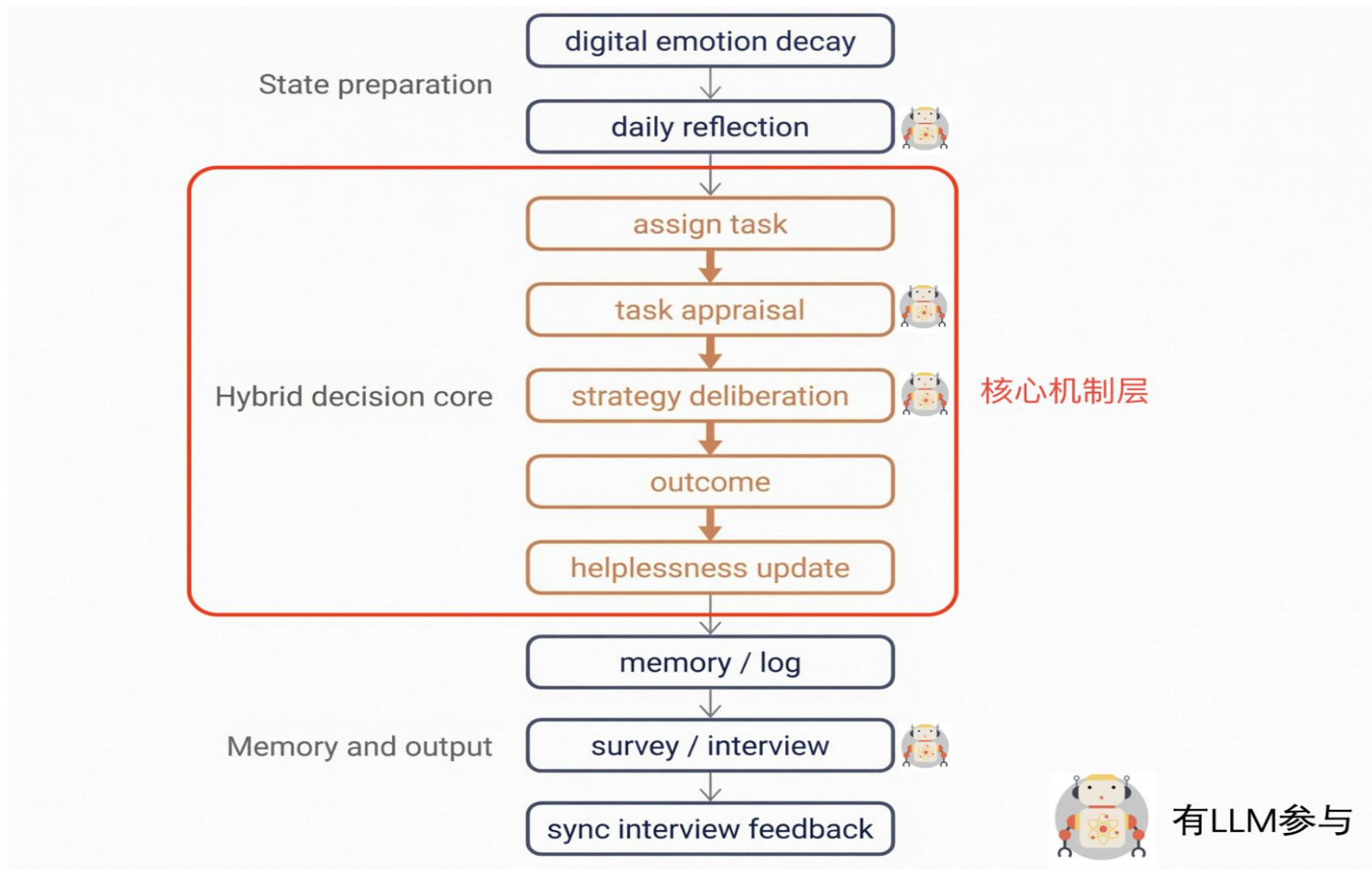
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  "persona": "谨慎务实者",
  "digital_experience": 0.2,
  "vision_limit": 0.6,
  "past_fraud_experience": 0.4,
  "background_story": "李春梅, 67岁, 退休后主要在社区活动, 平时喜欢去公园散步、参加社区合唱和健康讲座。她有一部用了多年的旧款手机, 日常只会接打电话和发微信, 偶尔刷短视频。视力下降让她对小字号和复杂页面特别不安。遇到验证码、弹窗或多步确认时容易焦虑, 常需要家人远程指导。她重视安全与省钱, 听到“优惠”“领奖”会本能警惕。"
},
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  "persona": "社区纽带",
  "digital_experience": 0.4,
  "vision_limit": 0.3,
  "past_fraud_experience": 0.1,
  "background_story": "周德荣, 71岁, 长期担任楼栋志愿者, 熟悉社区事务, 邻里有事常找他帮忙协调。他会用手机支付、网约车和挂号, 但对频繁弹窗、默认勾选和验证码输入非常不耐烦。过去帮邻居处理过几次误扣费问题, 因此对“看不清的按钮”和“套路流程”很反感。遇到系统更新或页面改版时, 他往往选择求助子女或直接放弃。"
},
{
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  "name": "王秀兰",
  "gender": "女",
  "age": 74,
  "education": "小学",
  "occupation": "退休",
  "persona": "保守谨慎者",
  "digital_experience": 0.1,
  "vision_limit": 0.7,
  "past_fraud_experience": 0.6,
  "background_story": "王秀兰, 74岁, 独居, 儿女在外地工作。几年前接到过“冒充客服退款”的电话, 虽然未被骗但受惊不小, 从此对线上支付极度敏感。她能用微信聊天和看新闻, 但涉及转账、验证码或银行卡绑定就会紧张, 常请求家人协助或干脆不用。视力不好使她对小字提示与复杂流程更不信任, 宁愿去线下窗口办理。"
```



DEMO流程



Agent核心机制示意图



Agent可能的演进过程

阶段一：初期受挫

>_ Action: Attempt_Back → Trigger_Ad

Helplessness: 0.1  强

“这个广告图标和返回键太像了，下次我要更仔细一点。”

阶段二：认知震荡

>_ Action: Retry_Pwd_x3 → Failed

Helplessness: 0.5  动摇

“输入三遍都不对……是我记错了，还是我真的看不清这些字？”

阶段三：彻底无助

>_ Action: Encounter_Popup → Exit

Helplessness: 0.9  回避

“又来了。我怎么点都是错的，我这种老头子根本不适合用智能手机。不添麻烦了。”

DEMO结果

Table 1: Paired effects vs baseline in vertical layout (Stage 6, 10-day, 3-seed

Contrast vs baseline	Metric	Mean Δ	95% CI
High Friction + Low Assist	Attempt rate	-0.349	[-0.390, -0.301]
	Neg. share	+0.449	[+0.418, +0.472]
	Helplessness	+18.614	[+16.829, +20.143]
	Trust	-19.783	[-20.425, -19.400]
	Avoidance	+7.167	[+4.850, +8.350]
High Friction + High Assist	Attempt rate	-0.116	[-0.144, -0.078]
	Neg. share	+0.195	[+0.171, +0.219]
	Helplessness	+14.645	[+9.296, +18.175]
	Trust	-22.017	[-29.950, -14.575]
	Avoidance	+12.033	[+4.325, +17.650]
Low Friction + High Assist	Attempt rate	+0.109	[+0.093, +0.133]
	Neg. share	-0.170	[-0.181, -0.155]
	Helplessness	-27.158	[-30.085, -24.262]
	Trust	+40.083	[+34.625, +44.000]
	Avoidance	-35.408	[-41.475, -30.375]

A

小幅改善整体趋稳

B

高摩擦会显著拉低表现

C

高辅助难抵高摩擦损害

初步方向与理论预期一致。

文献支持(以helplessness更新为例)

不应该是原本的数字摩擦带来的失败次数越多就越无助，而是应该由agent觉得自己做了也没用的程度来推动无助上升	事件结果不应该做为helplessness更新的主要参数	Seligman, M. E. P., and Maier, S. F. 1967. Failure to Escape Traumatic Shock. Abramson, L. Y., Seligman, M. E. P., and Teasdale, J. D. 1978. Learned Helplessness in Humans: Critique and Reformulation. Maier, S. F., and Seligman, M. E. P. 2016. Learned Helplessness at Fifty: Insights From Neuroscience.” <i>Psychological Review</i> Tafet, G. E., and Ortiz Alonso, F. J. 2025. Learned Helplessness and Learned Controllability: From Neurobiology to Cognitive, Emotional and Behavioral Neurosciences.
数字摩擦的失败不应该直接提升helplessness，而是先拉低效能感和控制感，再把人推向无助。	把自我效能感和自我的控制感放在agent的判断层中（这个还有待考虑到底怎么建模和放置），低效能、低控制感应当放大负面事件对helplessness的增幅，而相反的高效能、高控制感会起缓冲作用。 在代码中效能感和控制感也就是作为联系失败和helplessness的中间层	Bandura, A. 1977. Self-Efficacy: Toward a Unifying Theory of Behavioral Change. An, J., Zhu, X., Wan, K., Xiang, Z., Shi, Z., An, J., et al. 2024. Older Adults' Self-Perception, Technology Anxiety, and Intention to Use Digital Public Services.（这篇主要是tech anxiety，但我感觉好像也差不多） Sun, Y., Zhang, R., Hu, D., Wang, Y., Wen, C., et al. 2025. Factors That Influence Technophobia in Chinese Older Patients With Ischemic Stroke: A Cross-Sectional Survey.(技术恐惧) Song, X., Li, S., Dong, L., Li, Y., Yang, X., Bao, J., Liao, S., Xi, Y., and Guo, G. 2025. Technophobia in Digital Health Contexts: A Systematic Review and Meta-Analysis With a Focus on Older Adults(恐惧)
老人不做/回避数字行为的原因需要区分，不全因为无助，也有怕风险和觉得无用等因素	把事件结果的avoid分成helpless_avoid, risk_avoid和low_value_avoid。 只有因为无助而产生的回避才应该显著抬高helplessness，并影响memory	Davis, F. D. 1989. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. Venkatesh, V., Morris, M. G., Davis, G. B., and Davis, F. D. 2003. User Acceptance of Information Technology: Toward a Unified View. Jokisch, M. R., Schmidt, L. I., and Doh, M. 2022. Acceptance of Digital Health Services Among Older Adults: Findings on Perceived Usefulness, Self-Efficacy, Privacy Concerns, ICT Knowledge, and Support Seeking.
区分出老人真正依靠自己做成、而且感到可控的数字事件。这种事件才会形成长期的对于无助的抵抗，而不是简单的成功次数越多越好。	增加对于可控成功的记忆。 只有agent真正自己学会了、并且觉得这件事是自己能掌控的，才在以后更不容易陷入无助。 如果在support中是被别人完全帮着做完、自己并没有真正掌握的，对于无助的抵抗度应该比前者更低	Niu, H.-J., Li, M.-H., Hsieh, F.-Y., Yu, C.-C., and Lin, C.-T. 2026. From Digital Anxiety to Empowerment in Older Adults: Cross-Sectional Survey Study on Psychosocial Drivers of Digital Literacy. Lu, S. Y., Yoon, S., Yee, W. Q., Ngiam, N. H. W., Ng, K. Y. Y., and Low, L. L. 2024. Experiences of a Community-Based Digital Intervention Among Older People Living in a Low-Income Neighborhood: Qualitative Study
support不应该直接降低helplessness，而是通过提高效能感和控制感等中间层来影响helplessness	support应该作用到自我效能、可控感这类中间层，再间接影响 helplessness	Zhu, Y., Wang, X., Zhang, X., Li, Y., and Chen, Y. 2025. The Mechanism of Digital Feedback on Health Information Anxiety Among Older Adults: Information Processing Self-Efficacy as a Mediating Variable. Emmesjö, L., Hallgren, J., and Gillsjö, C. 2025. Older Adults’ Digital Technology Experiences: A Qualitative Study Jokisch, M. R., Schmidt, L. I., and Doh, M. 2022. Acceptance of Digital Health Services Among Older Adults: Findings on Perceived Usefulness, Self-Efficacy, Privacy Concerns, ICT Knowledge, and Support Seeking

项目预期产出

01 | 机制模型

形成 1 个老年人数字抵触动态机制模型

尝试通过数字摩擦 - 主观不可控感 - 自我效能下降 - 回避 - 数字抵触这一主要心理链条，构造可解释的老年人数字抵触形成与演进的机制模型

03 | 论文产出

预期完成1篇面向CCF-B/C投稿的论文初稿

。

02 | 实验框架

完成1套基于Agentsociety的，在平行世界下可对照的，由LLM Agent模拟的老年人在数字任务下抵触演化的实验

同时考虑将其制作为一个新的block提交给Agentsociety

请老师批评指正